

# Universal Derivation of All Physical Constants from the Fine-Structure Constant and Planck Length

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## Abstract

This document shows how the physical constants treated here can be calculated in SI units from two experimental base quantities together with the SI definitions (and, for atomic quantities, the electron mass) - the dimensionless fine-structure constant  $\alpha = 1/137.036$  and the Planck length  $\ell_P = 1.616255 \times 10^{-35}$  m. Additionally, the value of the characteristic energy  $E_0$  in T0 theory is discussed: the geometric mean of the CODATA electron and muon masses gives  $E_0 = 7.3486$  MeV; the value  $E_0 = 7.398$  MeV used for  $\alpha$  is interpreted here as the geometric mean of corrected masses. Two common circularity objections are discussed. The calculation reduces the number of natural constants that must be entered independently to two experimental values plus SI conventions, and the T0 raw values reproduce mass ratios without correction to about 0.5 %.

**Note on versions.** The German version of this document is more extensive than this English one. Earlier revisions shortened or changed parts of the English version. Where the two differ, the more extensive German version applies.

**Status markers.** Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

## .1 Introduction and Basic Principle

### The Minimal Principle of Physics

In modern physics, about 30 different natural constants appear to need independent experimental determination. This work shows that, together with the SI definitions, the constants treated here can be calculated from **two experimental values**:

### Fundamental Input Data

- **Fine-structure constant:**  $\alpha = \frac{1}{137.035999084}$  (dimensionless)
- **Planck length:**  $\ell_P = 1.616255 \times 10^{-35} \text{ m}$

### SI Base Definitions

Additionally, we use the modern SI base definitions (since 2019):

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m} \quad (\text{by definition}) \quad (1)$$

$$e = 1.602176634 \times 10^{-19} \text{ C} \quad (\text{exact definition}) \quad (2)$$

$$k_B = 1.380649 \times 10^{-23} \text{ J/K} \quad (\text{exact definition}) \quad (3)$$

$$N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1} \quad (\text{exact definition}) \quad (4)$$

## .2 Derivation of Fundamental Constants

### Speed of Light $c$

The speed of light follows from the relationship between Planck units. Since the Planck length is defined as:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \quad (5)$$

and all Planck units are interconnected through  $\hbar$ ,  $G$  and  $c$ , dimensional analysis yields:

### Speed of Light

$$c = 2.99792458 \times 10^8 \text{ m/s} \quad (6)$$

### Vacuum Permittivity $\epsilon_0$

From the Maxwell relation  $\mu_0 \epsilon_0 = 1/c^2$  follows:

$$\epsilon_0 = \frac{1}{\mu_0 c^2} = \frac{1}{4\pi \times 10^{-7} \times (2.99792458 \times 10^8)^2} \quad (7)$$

### Vacuum Permittivity

$$\epsilon_0 = 8.854187817 \times 10^{-12} \text{ F/m} \quad (8)$$

### Reduced Planck Constant $\hbar$

The fine-structure constant is defined as:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \quad (9)$$

Solving for  $\hbar$ :

$$\hbar = \frac{e^2}{4\pi\epsilon_0 c \alpha} \quad (10)$$

Substituting known values:

$$\hbar = \frac{(1.602176634 \times 10^{-19})^2}{4\pi \times 8.854187817 \times 10^{-12} \times 2.99792458 \times 10^8 \times \frac{1}{137.035999084}} \quad (11)$$

#### Reduced Planck Constant

$$\hbar = 1.054571817 \times 10^{-34} \text{ J}\cdot\text{s} \quad (12)$$

### Gravitational Constant G

From the definition of the Planck length follows:

$$G = \frac{\ell_P^2 c^3}{\hbar} \quad (13)$$

Substituting calculated values:

$$G = \frac{(1.616255 \times 10^{-35})^2 \times (2.99792458 \times 10^8)^3}{1.054571817 \times 10^{-34}} \quad (14)$$

#### Gravitational Constant

$$G = 6.67430 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2) \quad (15)$$

## .3 Complete Planck Units

With  $\hbar$ ,  $c$  and  $G$ , all Planck units can be calculated:

### Planck Time

$$t_P = \sqrt{\frac{\hbar G}{c^5}} = \frac{\ell_P}{c} = 5.391247 \times 10^{-44} \text{ s} \quad (16)$$

### Planck Mass

$$m_P = \sqrt{\frac{\hbar c}{G}} = 2.176434 \times 10^{-8} \text{ kg} \quad (17)$$

### Planck Energy

$$E_P = m_P c^2 = \sqrt{\frac{\hbar c^5}{G}} = 1.956082 \times 10^9 \text{ J} = 1.220890 \times 10^{19} \text{ GeV} \quad (18)$$

### Planck Temperature

$$T_P = \frac{E_P}{k_B} = \frac{m_P c^2}{k_B} = 1.416784 \times 10^{32} \text{ K} \quad (19)$$

## .4 Atomic and Molecular Constants

### Classical Electron Radius

With the electron mass  $m_e = 9.1093837015 \times 10^{-31}$  kg:

$$r_e = \frac{e^2}{4\pi\epsilon_0 m_e c^2} = \frac{\alpha \hbar}{m_e c} = 2.817940 \times 10^{-15} \text{ m} \quad (20)$$

### Compton Wavelength of the Electron

$$\lambda_{C,e} = \frac{h}{m_e c} = \frac{2\pi \hbar}{m_e c} = 2.426310 \times 10^{-12} \text{ m} \quad (21)$$

### Bohr Radius

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} = \frac{\hbar}{m_e c \alpha} = 5.291772 \times 10^{-11} \text{ m} \quad (22)$$

### Rydberg Constant

$$R_\infty = \frac{\alpha^2 m_e c}{2h} = \frac{\alpha^2 m_e c}{4\pi \hbar} = 1.097373 \times 10^7 \text{ m}^{-1} \quad (23)$$

## .5 Thermodynamic Constants

### Stefan-Boltzmann Constant

$$\sigma = \frac{2\pi^5 k_B^4}{15h^3 c^2} = \frac{2\pi^5 k_B^4}{15(2\pi \hbar)^3 c^2} = 5.670374419 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K}^4) \quad (24)$$

### Wien's Displacement Law Constant

$$b = \frac{hc}{k_B} \times \frac{1}{4.965114231} = 2.897771955 \times 10^{-3} \text{ m} \cdot \text{K} \quad (25)$$

## .6 Dimensional Analysis and Verification

### Consistency Check of the Fine-Structure Constant

$$[\alpha] = \frac{[e^2]}{[\epsilon_0][\hbar][c]} \quad (26)$$

$$= \frac{[\text{C}^2]}{[\text{F}/\text{m}][\text{J} \cdot \text{s}][\text{m}/\text{s}]} \quad (27)$$

$$= \frac{[\text{C}^2]}{[\text{C}^2 \cdot \text{s}^2 / (\text{kg} \cdot \text{m}^3)][\text{J} \cdot \text{s}][\text{m}/\text{s}]} \quad (28)$$

$$= \frac{[\text{C}^2]}{[\text{C}^2 / (\text{kg} \cdot \text{m}^2 / \text{s}^2)]} \quad (29)$$

$$= [1] \quad \checkmark \quad (30)$$

## Consistency Check of the Gravitational Constant

$$[G] = \frac{[\ell_p^2][c^3]}{[\hbar]} \quad (31)$$

$$= \frac{[m^2][m^3/s^3]}{[J \cdot s]} \quad (32)$$

$$= \frac{[m^5/s^3]}{[kg \cdot m^2/s^2 \cdot s]} \quad (33)$$

$$= \frac{[m^5/s^3]}{[kg \cdot m^2/s^3]} \quad (34)$$

$$= [m^3/(kg \cdot s^2)] \quad \checkmark \quad (35)$$

## Consistency Check of $\hbar$

$$[\hbar] = \frac{[e^2]}{[\varepsilon_0][c][\alpha]} \quad (36)$$

$$= \frac{[C^2]}{[F/m][m/s][1]} \quad (37)$$

$$= \frac{[C^2]}{[C^2 \cdot s/(kg \cdot m^3)][m/s]} \quad (38)$$

$$= \frac{[C^2 \cdot kg \cdot m^3]}{[C^2 \cdot s \cdot m]} \quad (39)$$

$$= [kg \cdot m^2/s] = [J \cdot s] \quad \checkmark \quad (40)$$

## .7 The Characteristic Energy E\_0 and T0 Theory

### Definition of the Characteristic Energy

#### Basic Definition

The fundamental definition of the characteristic energy is:

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (41)$$

This is **not a derivation** and **not a fit** – it is the mathematical definition of the geometric mean of two masses.

### Numerical Evaluation with Different Precision Levels

#### Level 1: Rounded Standard Values

With the often cited rounded masses:

$$m_e = 0.511 \text{ MeV} \quad (42)$$

$$m_\mu = 105.658 \text{ MeV} \quad (43)$$

$$E_0^{(1)} = \sqrt{0.511 \times 105.658} = \sqrt{53.99} = 7.348 \text{ MeV} \quad (44)$$

## Level 2: CODATA 2018 Precision Values

With the precise experimental masses:

$$m_e = 0.510,998,946,1 \text{ MeV} \quad (45)$$

$$m_\mu = 105.658,374,5 \text{ MeV} \quad (46)$$

$$E_0^{(2)} = \sqrt{0.5109989461 \times 105.6583745} = 7.348,566 \text{ MeV} \quad (47)$$

## Level 3: The Value $E_0 = 7.398 \text{ MeV}$

### Critical Question

**Is  $E_0 = 7.398 \text{ MeV}$  a fitted parameter?**

**Answer in this document:** It is not treated as a free fit parameter.

$E_0 = 7.398 \text{ MeV}$  is interpreted here as the geometric mean of refined mass values that are meant to include corrections (QED, weak interaction). These corrections are not derived individually in this document; the geometric mean of the CODATA masses themselves is  $7.3486 \text{ MeV}$  (Level 2).

## Precise Fine-Structure Constant Calculation

The dimensionally correct formula:

$$\alpha = \xi \cdot \frac{E_0^2}{(1 \text{ MeV})^2} \quad (48)$$

where:

- $\xi = \frac{4}{3} \times 10^{-4} = 1.333\bar{3} \times 10^{-4}$  (exact)
- $(1 \text{ MeV})^2$  is the normalization energy for dimensionless calculation

## Comparison of Calculation Accuracy

| $E_0$ Value                 | Source           | $\alpha_{\text{TO}}^{-1}$ | Deviation        |
|-----------------------------|------------------|---------------------------|------------------|
| 7.348 MeV                   | Rounded masses   | 139.15                    | 1.5%             |
| 7.348,566 MeV               | CODATA exact     | 139.07                    | 1.4%             |
| 7.398 MeV                   | <b>Optimized</b> | <b>137.038</b>            | <b>0.0014%</b>   |
| <b>Experiment (CODATA):</b> |                  | <b>137.035999084</b>      | <b>Reference</b> |

**Table 1:** Comparison of calculation accuracy for different  $E_0$  values

**Detailed Calculation with  $E_0 = 7.398 \text{ MeV}$** 

$$E_0^2 = (7.398)^2 = 54.7303 \text{ MeV}^2 \quad (49)$$

$$\frac{E_0^2}{(1 \text{ MeV})^2} = 54.7303 \quad (50)$$

$$\alpha = 1.333\bar{3} \times 10^{-4} \times 54.7303 \quad (51)$$

$$= 7.297 \times 10^{-3} \quad (52)$$

$$\alpha^{-1} = 137.038 \quad (53)$$

**Comparison with Experiment**

**T0 Prediction:**  $\alpha^{-1} = 137.038$

**Experiment:**  $\alpha^{-1} = 137.035999084$

**Relative Deviation:**  $\frac{|137.038 - 137.036|}{137.036} = 0.0014\%$

**.8 On the Precision of the Value  $E_0 = 7.398 \text{ MeV}$** **Why  $E_0 = 7.398 \text{ MeV}$  Works Well**

According to the interpretation of this document, the value  $E_0 = 7.398 \text{ MeV}$  is meant to result from:

1. **Inclusion of QED corrections** in particle masses
2. **Incorporation of weak interaction effects**
3. **Geometric mean calculation** with full precision
4. **Consistency** with T0 geometry  $\xi = \frac{4}{3} \times 10^{-4}$

**The Mathematical Justification****Geometric Interpretation**

The geometric mean  $E_0 = \sqrt{m_e \cdot m_\mu}$  is the natural energy scale between electron and muon.

On a logarithmic scale,  $E_0$  lies precisely in the middle:

$$\log(E_0) = \frac{\log(m_e) + \log(m_\mu)}{2} \quad (54)$$

This is the **characteristic energy** of the first two lepton generations.

## .9 Comparison with Alternative Approaches

### Estimation with T0-Calculated Masses

If the particle masses themselves were calculated from T0 theory:

$$m_e^{T0} = 0.511,000 \text{ MeV} \quad (\text{theoretical}) \quad (55)$$

$$m_\mu^{T0} = 105.658,000 \text{ MeV} \quad (\text{theoretical}) \quad (56)$$

$$E_0^{T0} = \sqrt{0.511000 \times 105.658000} = 72.868 \text{ MeV} \quad (57)$$

**Problem:** This calculation is obviously flawed ( $E_0 = 72.868 \text{ MeV}$  is much too large).

### Correct Interpretation

The correct approach is:

1. Use **experimental masses** as input
2. Calculate **geometric mean** exactly
3. Use **T0 geometry**  $\xi$  as theoretical parameter
4. Check **fine-structure constant** as output

## .10 Dimensional Consistency of the E\_0 Formula

### Correct Dimensionless Formulation

The formula:

$$\alpha = \xi \cdot \frac{E_0^2}{(1 \text{ MeV})^2} \quad (58)$$

is dimensionally consistent:

$$[\alpha] = [\xi] \cdot \frac{[E_0^2]}{[(1 \text{ MeV})^2]} \quad (59)$$

$$= [1] \cdot \frac{[\text{Energy}^2]}{[\text{Energy}^2]} \quad (60)$$

$$= [1] \quad \checkmark \quad (61)$$

### Alternative Notation

Equivalently can be written:

$$\frac{1}{\alpha} = \frac{(1 \text{ MeV})^2}{\xi \cdot E_0^2} = \frac{1}{\xi \cdot 54.73} = \frac{1}{1.333 \times 10^{-4} \times 54.73} = 137.038 \quad (62)$$

## .11 On the Circularity Objections

### The Apparent Circularity Objections

#### Common Criticisms

**Objection 1:** The Planck length  $\ell_P$  is already defined via the gravitational constant  $G$ :

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \quad (63)$$

Therefore, it is circular to derive  $G$  from  $\ell_P$ .

**Objection 2:** The speed of light  $c$  is calculated from  $\mu_0$  and  $\varepsilon_0$ :

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \quad (64)$$

But  $\varepsilon_0$  is calculated from  $c$  - that is circular.

### Resolution of the Apparent Circularity

#### The Structure of SI Definitions (since 2019)

##### Modern SI Base

Since the SI reform in 2019, the following quantities are **exactly defined**:

$$c = 299792458 \text{ m/s} \quad (\text{exact definition}) \quad (65)$$

$$e = 1.602176634 \times 10^{-19} \text{ C} \quad (\text{exact definition}) \quad (66)$$

$$\hbar = 1.054571817 \times 10^{-34} \text{ J}\cdot\text{s} \quad (\text{exact definition}) \quad (67)$$

$$k_B = 1.380649 \times 10^{-23} \text{ J/K} \quad (\text{exact definition}) \quad (68)$$

Only  $\mu_0$  is still calculated:  $\mu_0 = \frac{4\pi \times 10^{-7}}{\text{defined}}$

### Corrected Hierarchy with Modern SI

The actual derivation is therefore:

$$\text{Given (experimental): } \alpha, \ell_P \quad (69)$$

$$\text{Defined (SI 2019): } c, e, \hbar, k_B \quad (70)$$

$$\text{Calculated: } \varepsilon_0 = \frac{e^2}{4\pi\hbar c\alpha} \quad (71)$$

$$\mu_0 = \frac{1}{\varepsilon_0 c^2} \quad (72)$$

$$G = \frac{\ell_P^2 c^3}{\hbar} \quad (73)$$

**Result:** For  $c$ ,  $\varepsilon_0$  and  $\mu_0$  there is no circularity, since  $c$  and  $\hbar$  are directly defined. For  $G$  it should be noted that, in practice,  $\ell_P$  has so far been determined from the measured  $G$ .

### $\ell_P$ is Only ONE Possible Length Scale

The Planck length is not the only fundamental length scale. One could equally well use:

$$L_1 = 2.5 \times 10^{-35} \text{ m} \quad (\text{arbitrarily chosen}) \quad (74)$$

$$L_2 = 1.0 \times 10^{-35} \text{ m} \quad (\text{round number}) \quad (75)$$

$$L_3 = \pi \times 10^{-35} \text{ m} \quad (\text{with } \pi) \quad (76)$$

$$L_4 = e \times 10^{-35} \text{ m} \quad (\text{with } e) \quad (77)$$

### The Mathematics Works with ANY Length Scale

The general formula is:

$$G = \frac{L^2 \times c^3}{\hbar} \quad (78)$$

**Crucial:** Only with the specific length  $\ell_P = 1.616255 \times 10^{-35} \text{ m}$  does one obtain the correct experimental value of  $G$ .

### The SI Reference is What Matters

| Length Scale L                             | Calculated G                                                   | Status         |
|--------------------------------------------|----------------------------------------------------------------|----------------|
| $2.5 \times 10^{-35} \text{ m}$            | $1.04 \times 10^{-10} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$  | Wrong          |
| $1.0 \times 10^{-35} \text{ m}$            | $1.67 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$  | Wrong          |
| $\pi \times 10^{-35} \text{ m}$            | $1.64 \times 10^{-10} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$  | Wrong          |
| $\ell_P = 1.616 \times 10^{-35} \text{ m}$ | $6.674 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$ | <b>Correct</b> |

**Table 2:** G-values for different length scales

### The Hierarchy in the T0 Interpretation

#### Interpretation in the T0 Framework

In the T0 interpretation,  $\ell_P$  and  $G$  are both expressions of the same underlying geometry.

#### The order in this interpretation:

1. Fundamental 3D space geometry  $\rightarrow \xi = \frac{4}{3} \times 10^{-4}$
2. From this follows  $\ell_P$  as natural scale
3. From this follows  $G$  as emergent property
4. SI units provide the reference to human measures

### Possible Independent Approaches

#### Independent Measurement of $\ell_P$

The Planck length could in principle become accessible independently of  $G$  through **[S]** (no such measurement exists so far):

1. **Quantum gravity experiments:** Direct measurement of the minimal length scale

2. **Black hole Hawking radiation:**  $\ell_P$  determines the evaporation rate
3. **Cosmological observations:**  $\ell_P$  influences quantum fluctuations of inflation (cosmic sector, deliberately set aside in the current corpus, P39)
4. **High-energy scattering experiments:** At Planck energies,  $\ell_P$  becomes directly accessible

### Independent Measurement of $\alpha$

The fine-structure constant is measured through:

1. **Quantum Hall effect:**  $\alpha = \frac{e^2}{h} \times \frac{R_K}{Z_0}$
  2. **Anomalous magnetic moment:**  $\alpha$  from QED corrections
  3. **Atom interferometry:**  $\alpha$  from recoil measurements
  4. **Spectroscopy:**  $\alpha$  from hydrogen spectrum
- None of these methods uses  $G$  or  $\ell_P$ .

### Formal Check of the Order of Definitions

#### Definition Hierarchy

**Given:**  $\alpha$  (experimental),  $\ell_P$  (experimental) (79)

**Defined:**  $\mu_0$  (SI convention),  $e$  (SI convention) (80)

**Calculated:**  $c = f_1(\mu_0)$ ,  $\epsilon_0 = f_2(\mu_0, c)$  (81)

$\hbar = f_3(e, \epsilon_0, c, \alpha)$  (82)

$G = f_4(\ell_P, c, \hbar)$  (83)

**Each quantity depends only on previously defined quantities.**

#### Circularity Test

A circular argument exists if:

$$A \xrightarrow{\text{defined}} B \xrightarrow{\text{defined}} C \xrightarrow{\text{defined}} A \quad (84)$$

In our case:

$$\alpha, \ell_P \xrightarrow{\text{calculated}} \hbar \xrightarrow{\text{calculated}} G \leftrightarrow \alpha, \ell_P \quad (85)$$

**Result:** Within this chain of calculation there is no circularity.

## The Philosophical Argument

### Reference Scales are Necessary

#### Reference Scales

**All physics needs reference scales.**

Nature is dimensionally structured. To get from dimensionless relationships to measurable quantities, we need:

- An **energy scale** (from  $\alpha$ )
- A **length scale** (from  $\ell_P$ )
- **SI conventions** (human measures)

This holds for any dimensional physics, not only for T0 theory.

## .12 Further Considerations

### Connection to the T0 Model

Within the T0 model,  $\alpha$  and  $\ell_P$  could also be derived from geometric principles **[S]**:

$$\xi = \frac{4}{3} \times 10^{-4} \quad (\text{3D space geometry}) \quad (86)$$

$$\alpha = \xi \times E_0^2 \quad \text{with } E_0 = \sqrt{m_e \times m_\mu} \quad (87)$$

$$\ell_P = \xi \times \ell_{\text{fundamental}} \quad (88)$$

This would reduce the number of fundamental parameters to **one**, the geometric parameter  $\xi$  (plus motivated structural choices and SI anchors for unit conversion).